



Best Method for solving Recurrence

Answer Iteration Method.

Explanation:-

① The recurrence reduces the problem by 1 each time

② Since recurrence is linear, expanding step by step is the easiest

Solution of the Iteration

$$T(n) = T(n-1) + n$$

$$T(n) = T(n-1) + n$$

$$= T(n-2) + (n-1) + n$$

$$= T(n-3) + (n-2) + (n-1) + n$$

$$\text{to } \Rightarrow T(1) + 2 + 3 + \dots + n$$

Hence the sum

$$2 + 3 + \dots + n = \frac{n(n+1)}{2} - 1$$

$$\therefore T(n) = O(n^2)$$

$$\text{Time Complexity} = O(n^2)$$



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Using Master Theorem

$$T(n) = \alpha T(n/b) + f(n)$$

where $\alpha=3$ and $b=3$ $f(n)=n$

Hence computing

$$n \log_b \alpha = n \log_3 3 = n' = n$$

$$f(n) = n \log_b \alpha$$

$$T(n) = O(n \log n)$$

Using Characteristic Equation Method

$$T(n) = 2T(n-1) + 1$$

Solving the homogeneous part

$$T(n) = 2T(n-1)$$

Finding the characteristic eqn $r=2$

$$T_h(n) = C \cdot 2^n$$

Since RHS is 1 we assume

$$T_p = A$$

$$A = 2A + 1$$

$$\underline{\underline{A = -1}}$$

$$T(n) = C \cdot 2^n - 1$$

$$\text{Time complexity} = O(2^n)$$